

PRISM MODEL EVALUATION REPORT

gemma3:4b

Prompt Reliability Through Intelligent Semantic Multiplexing

Generated 19 August 2026

Abstract

This report evaluates gemma3:4b on 2 datasets totaling 400 questions, each posed under 5 paraphrased prompt conditions (P0-P4) to measure not only answer correctness but the stability of that correctness under prompt paraphrase. Mean prompt-response accuracy across datasets was 81.2%, with mean cross-prompt agreement of 95.8% and a mean prompt-invariant-incorrect rate of 10.2% -- questions answered the same way under every phrasing, but incorrectly. Section I reports per-dataset summary metrics, Section II reports the same metrics broken out by prompt condition, and the remaining sections present the corresponding figures.

Index Terms

large language models, benchmark evaluation, prompt robustness, cross-prompt agreement, answer consistency.

I. Summary Results

Table I reports, for each dataset, the question count and the headline metrics computed across all prompt conditions: prompt-response accuracy, conditional accuracy, answer recovery rate, instruction compliance rate, mean cross-prompt agreement, mean prompt sensitivity, the answer-unanimous rate, and the prompt-invariant-incorrect rate.

TABLE I
Per-Dataset Summary — gemma3:4b

Dataset	N	Prompt-Resp. Acc.	Cond. Acc.	Recovery	Compliance	Agreement	Sensitivity	Unanimous	Invariant- Wrong
Arc Challenge	200	73.8%	73.8%	100.0%	100.0%	94.3%	5.7%	79.0%	13.5%
Sciq	200	88.5%	88.6%	99.9%	99.9%	97.4%	2.6%	91.5%	7.0%

Note: consistency is not correctness. A model can answer identically across all five prompt phrasings while being consistently wrong — see the Invariant-Wrong column above.

II. Metrics by Prompt Condition

Table II breaks the Table I metrics down by dataset and prompt condition (P0, the unmodified baseline prompt, through P4), so that dataset-specific sensitivity to prompt phrasing is visible rather than averaged away.

TABLE II
Metrics by Prompt Condition — gemma3:4b

Dataset	Cond.	Accuracy	Cond. Acc.	Agreement	Sensitivity	Compliance	Recovery	Unknown
Arc Challenge	P0	75.0%	75.0%	94.3%	5.7%	100.0%	100.0%	0.0%
Arc Challenge	P1	74.0%	74.0%	94.3%	5.7%	100.0%	100.0%	0.0%
Arc Challenge	P2	74.5%	74.5%	94.3%	5.7%	100.0%	100.0%	0.0%
Arc Challenge	P3	73.5%	73.5%	94.3%	5.7%	100.0%	100.0%	0.0%
Arc Challenge	P4	72.0%	72.0%	94.3%	5.7%	100.0%	100.0%	0.0%
Sciq	P0	88.5%	88.5%	97.4%	2.6%	100.0%	100.0%	0.0%
Sciq	P1	88.5%	88.9%	97.4%	2.6%	99.5%	99.5%	0.5%
Sciq	P2	89.0%	89.0%	97.4%	2.6%	100.0%	100.0%	0.0%
Sciq	P3	87.5%	87.5%	97.4%	2.6%	100.0%	100.0%	0.0%
Sciq	P4	89.0%	89.0%	97.4%	2.6%	100.0%	100.0%	0.0%

III. Accuracy by Dataset

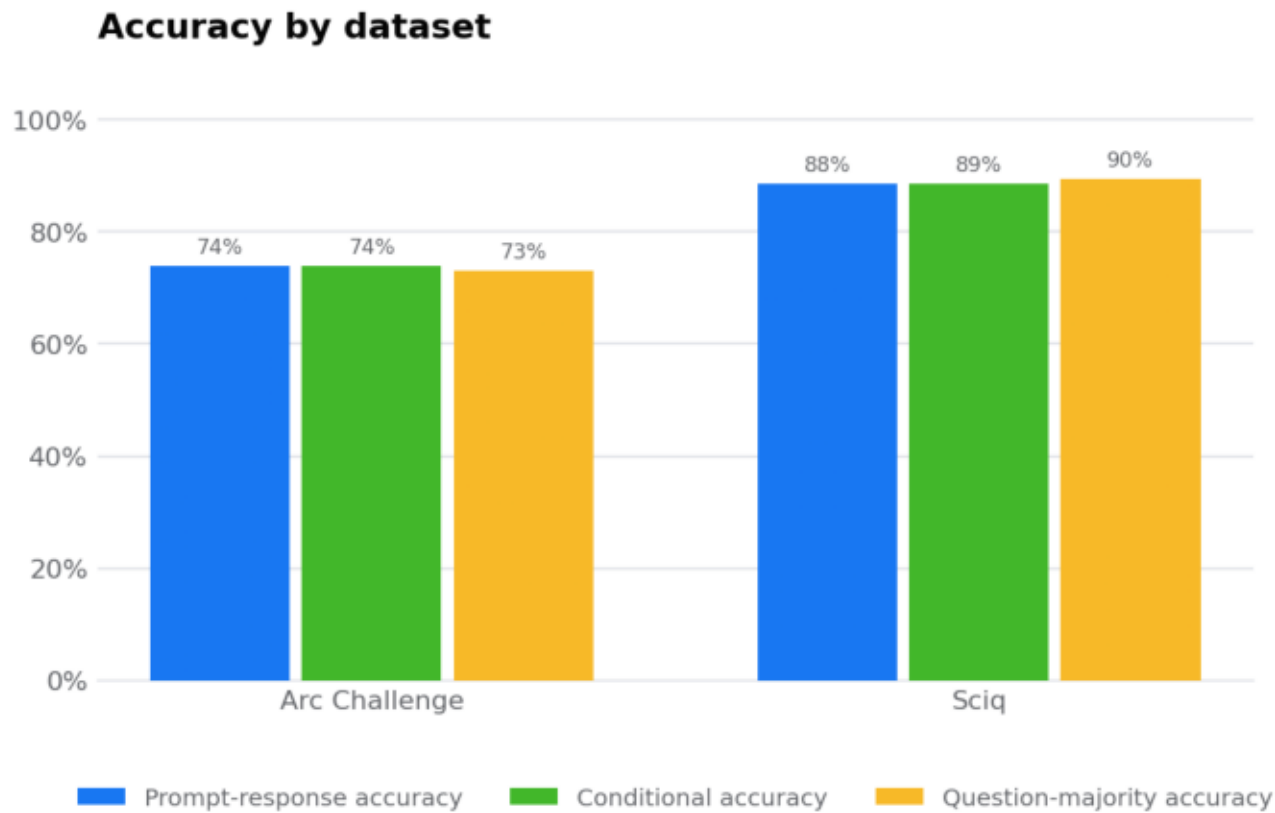


Fig. 1. Prompt-response accuracy, conditional accuracy, and question-majority accuracy for gemma3:4b on each evaluated dataset. Prompt-response accuracy is computed over every individual prompt response; conditional accuracy restricts to instruction-compliant responses; question-majority accuracy uses the majority answer across the five prompt phrasings for each question.

IV. Cross-Prompt Agreement vs. Prompt Sensitivity

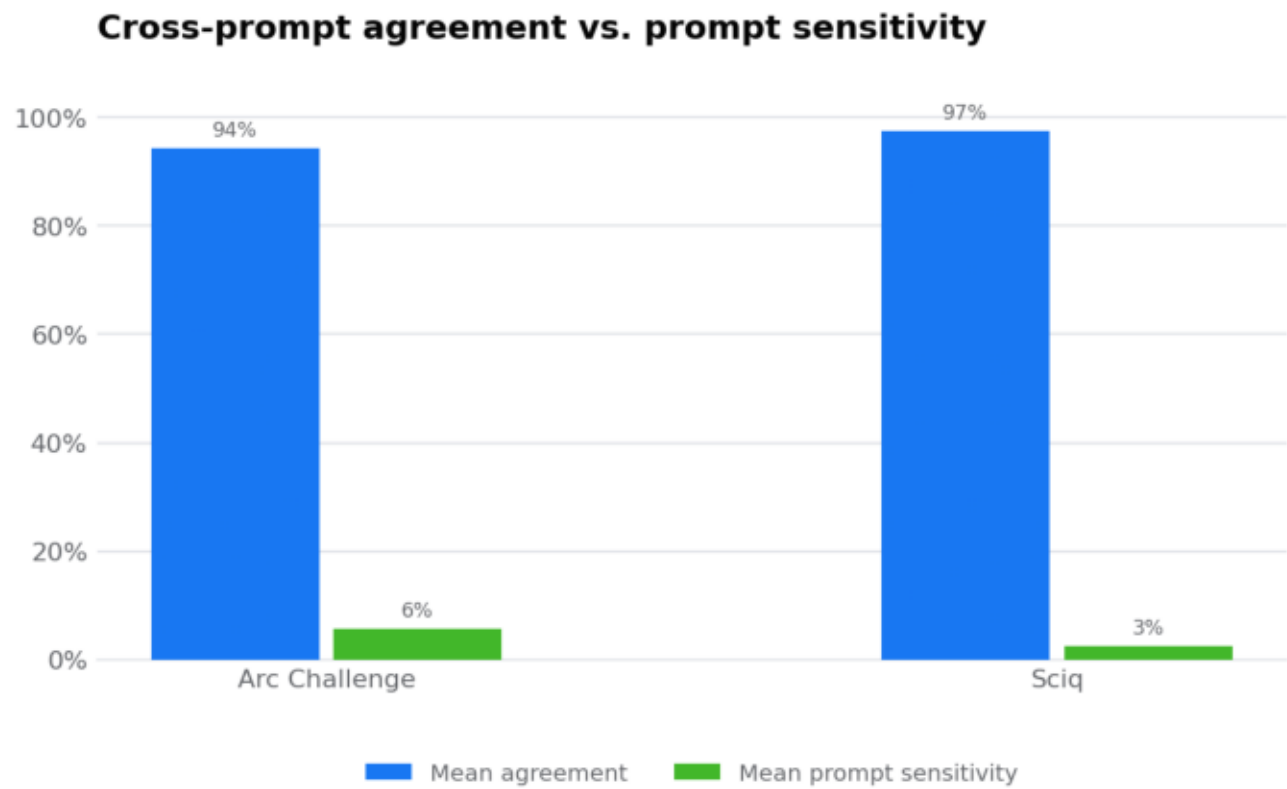


Fig. 2. Mean agreement (the rate at which gemma3:4b gives the same answer across prompt phrasings for a question) against mean prompt sensitivity (the rate at which its answer changes), by dataset. The two are complementary views of answer stability under paraphrase.

V. Unanimous Agreement vs. Unanimous-but-Wrong

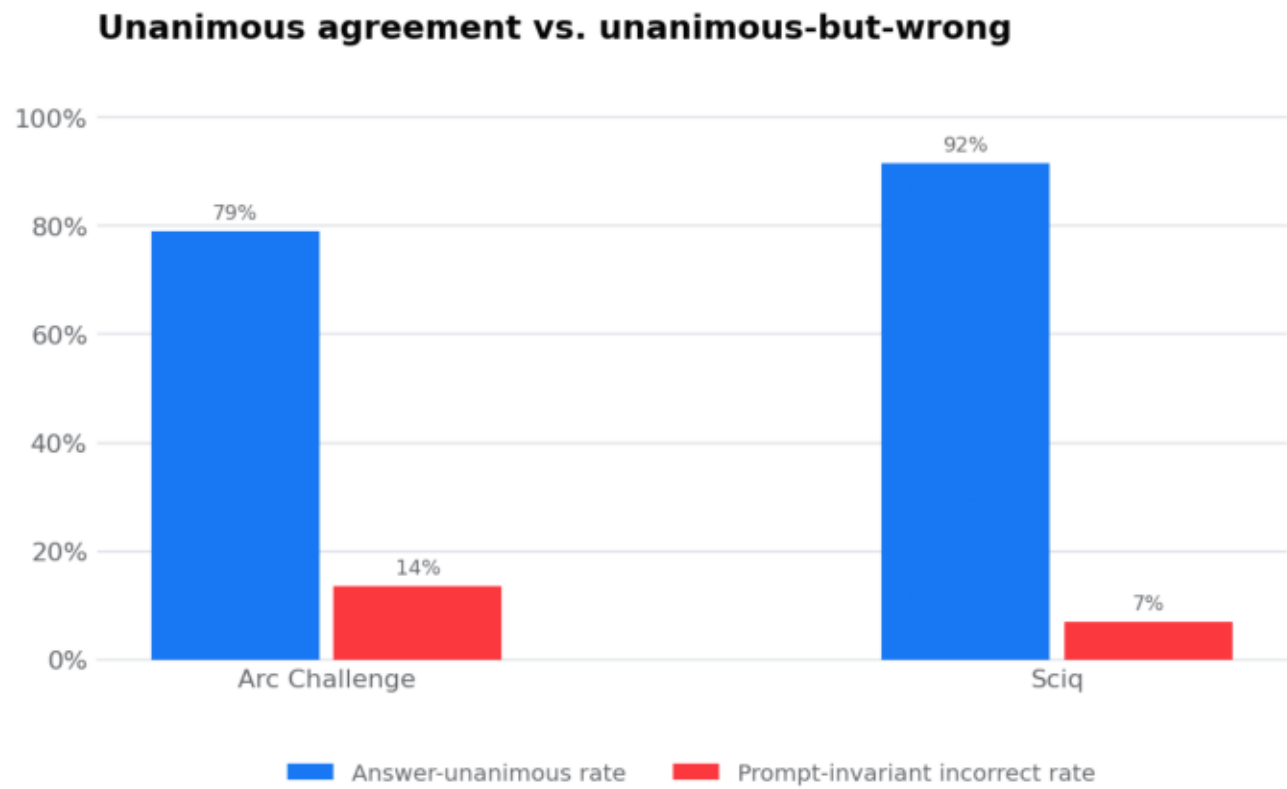


Fig. 3. The answer-unanimous rate (all five prompt phrasings yield the same answer) against the prompt-invariant incorrect rate (all five phrasings agree, but on the wrong answer). A model can be perfectly consistent while being consistently wrong; this figure isolates that failure mode.

VI. Response Quality by Dataset

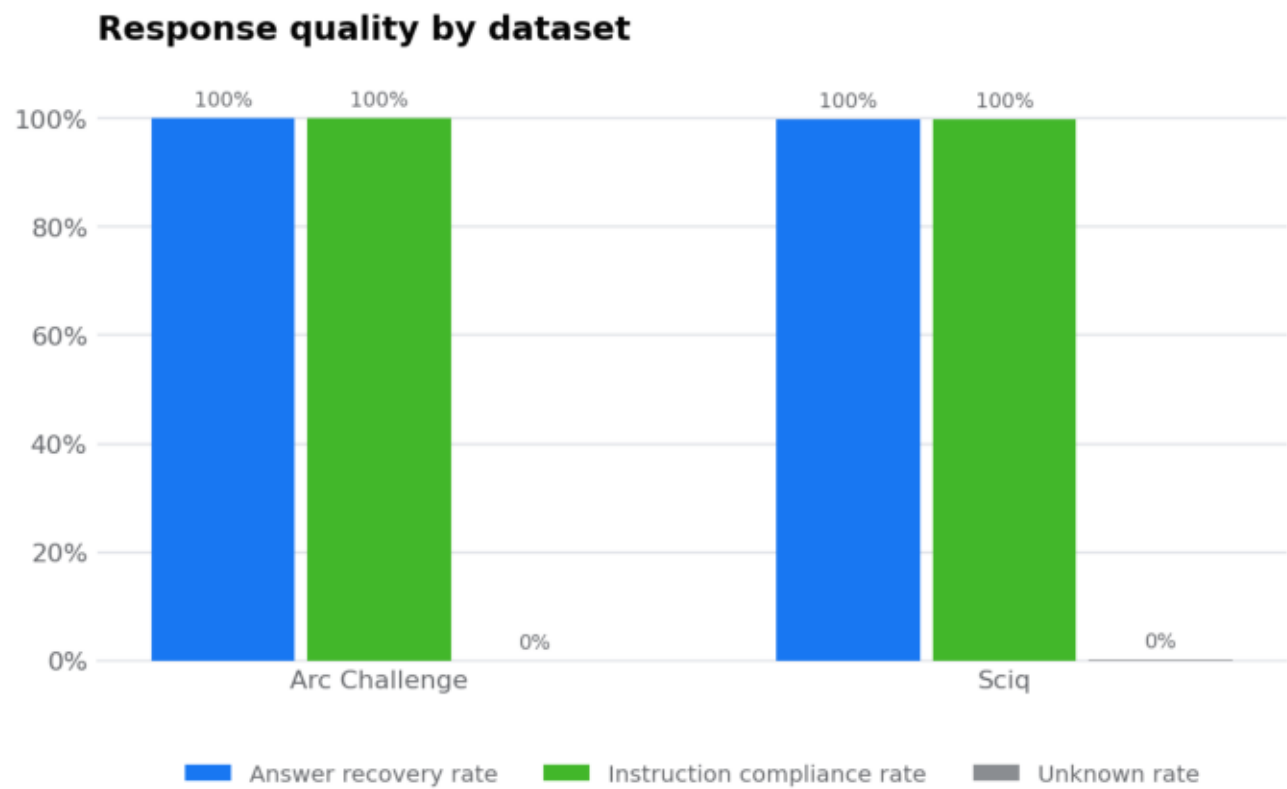


Fig. 4. Answer recovery rate, instruction compliance rate, and unknown rate for gemma3:4b on each dataset, characterizing how often a usable answer was produced and how often the requested output format was followed.

VII. Accuracy Across Prompt Conditions

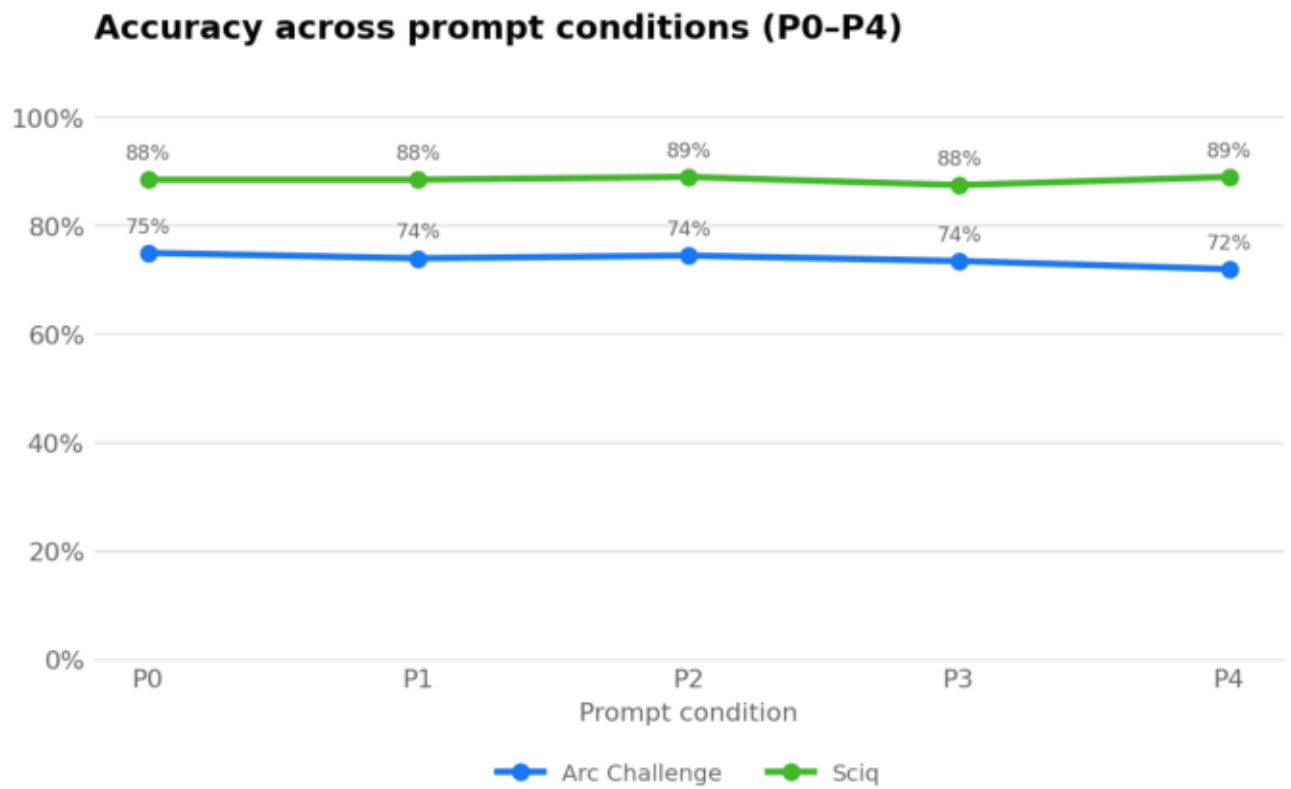


Fig. 5. Accuracy for gemma3:4b on each dataset as the prompt condition is varied from P0 (baseline) through P4, showing whether accuracy degrades under paraphrase or added instructions.

VIII. Accuracy by Prompt Condition

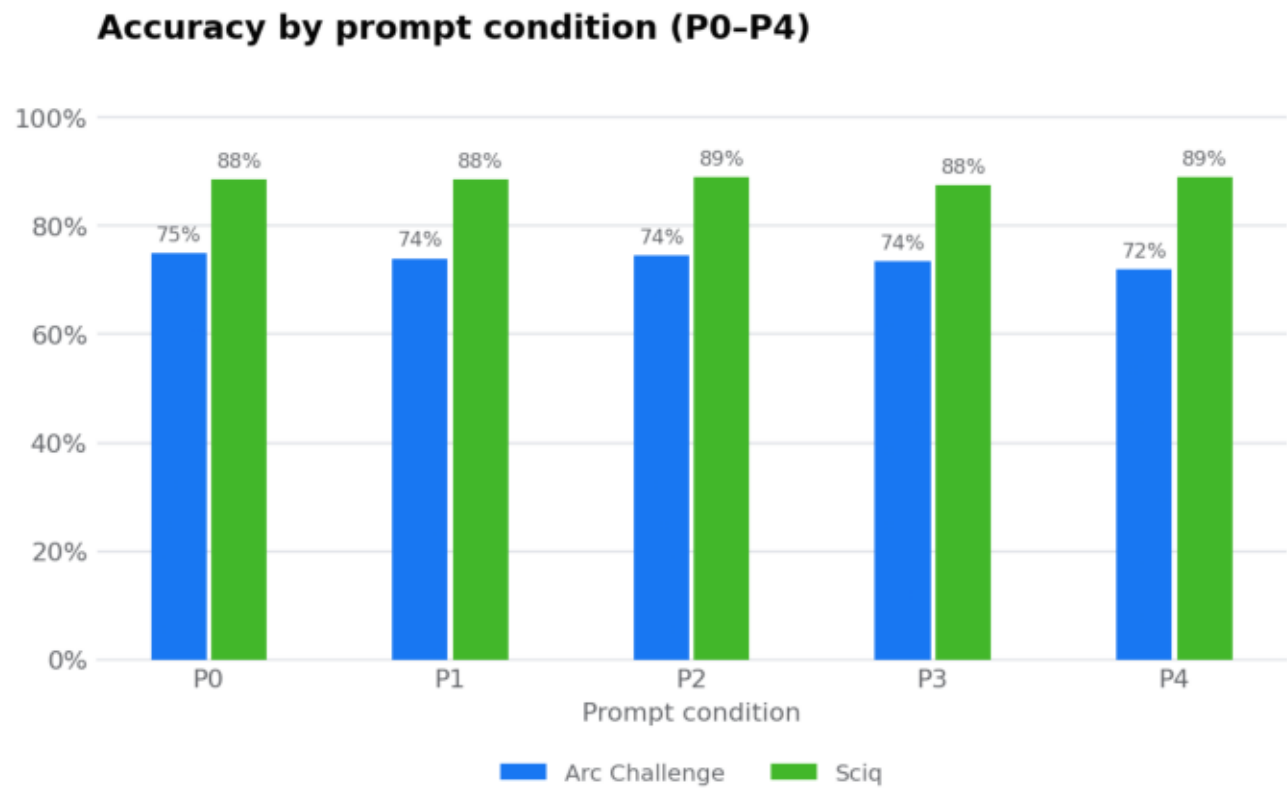


Fig. 6. Per-dataset accuracy at each prompt condition (P0-P4), shown as grouped bars for direct comparison across datasets at a given condition.

IX. Instruction Compliance by Prompt Condition

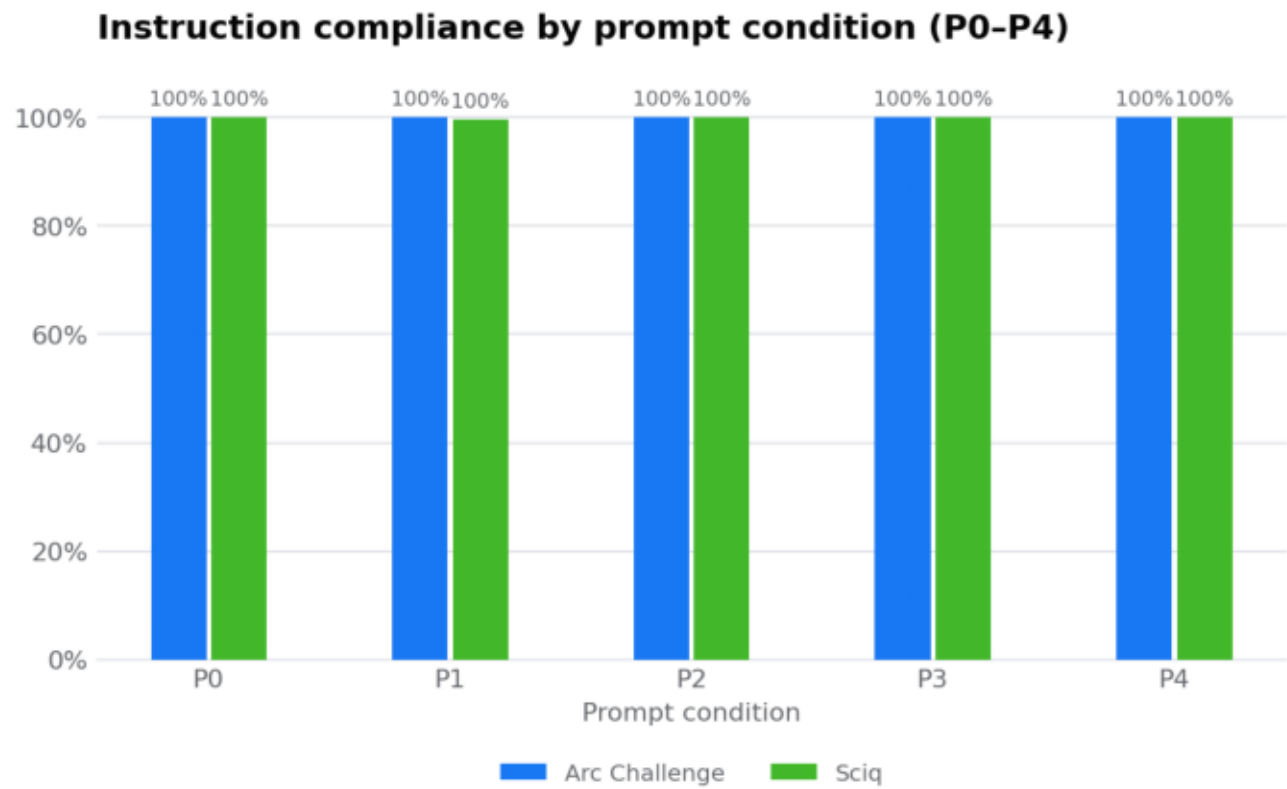


Fig. 7. The rate at which gemma3:4b followed the requested output instructions at each prompt condition, by dataset.

X. Prompt Sensitivity by Prompt Condition

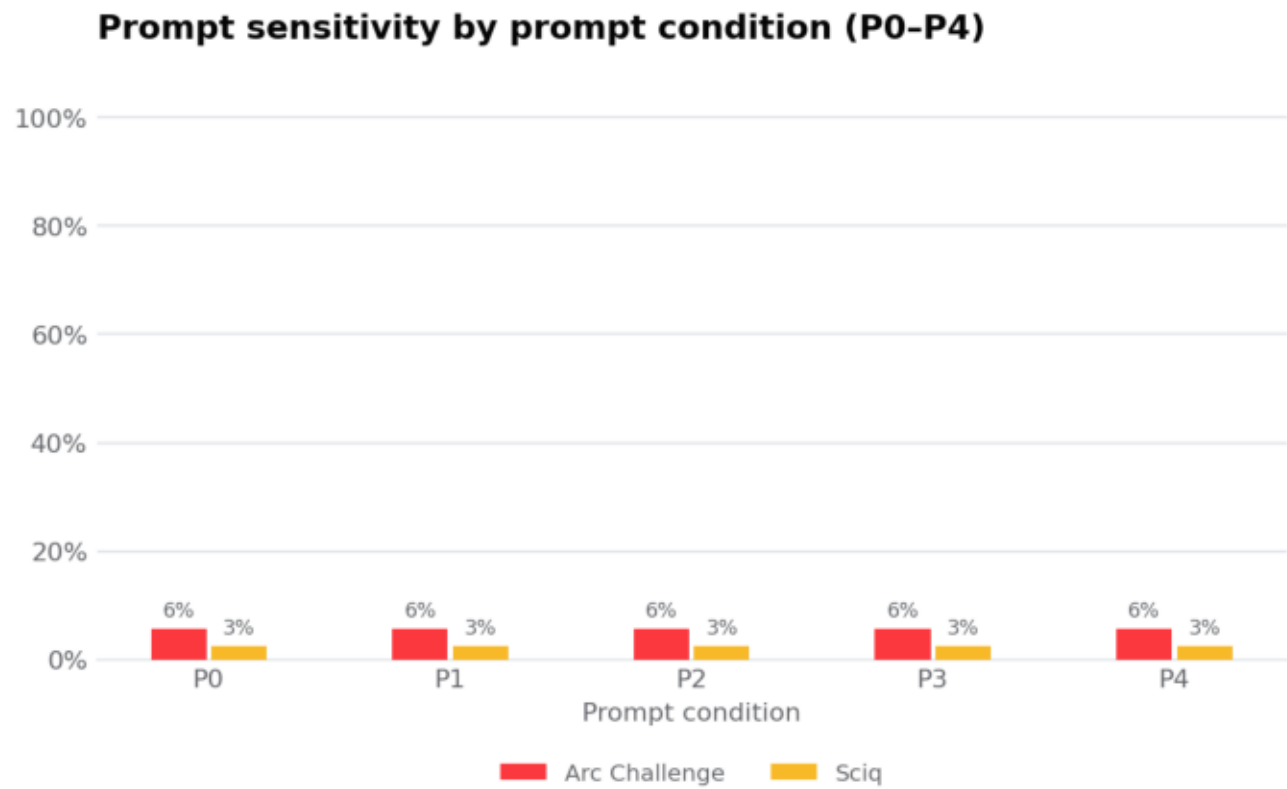


Fig. 8. The rate at which gemma3:4b's answer changed relative to its P0 baseline answer at each subsequent prompt condition, by dataset.